Chapter 14
Wiener Processes and Itô's Lemma †

Chapter 13 priced options without any continuous-time mathematics. It could do so because the binomial world has finitely many states, and finitely many states can be handled with algebra. Chapter 15 does the same job in continuous time, where the states form a continuum and algebra is no longer enough. This chapter supplies the language for that. A reader who wants the Black–Scholes–Merton formula and its use rather than its derivation can pass from Chapter 13 straight to Chapter 15 without loss of continuity. Chapters 15, 16, 19, and 22 do cite four results here worth taking even on a skip: the lognormal property (Section 14.9: the median grows at $\mu - \sigma^2/2$, more slowly than the mean, a gap behind common errors in performance reporting), the $\sigma/16$ rule (Section 14.6), the volatility of a ratio (Section 14.11) and the model's failures (Section 14.12). Chapters 20, 21, 28, and 30 assume the whole chapter.

Sections 14.1 to 14.4 build the process for a stock price, from the Markov property through Wiener processes to geometric Brownian motion; Sections 14.5 and 14.6 say what its two parameters mean. Sections 14.7 and 14.8 are Itô's lemma and why it delivers a hedge; Sections 14.9 and 14.10 are the lognormal property and the risk-neutral process. Sections 14.11 and 14.12 extend the model to two sources of uncertainty and state what it assumes, and Section 14.13 introduces fractional Brownian motion, which Chapter 21's rough volatility will need.

14.1 THE MARKOV PROPERTY

A **Markov process** is one in which only the present value of the variable is relevant for predicting the future. The path taken to get there carries no information.

Stock prices are usually assumed to be Markov. The motivation is **weak-form market efficiency**, the hypothesis that past prices cannot be used to predict future ones: if the history of prices contained information about future prices beyond what today's price contains, that information could be traded on, and trading on it would remove it. The assumption is not exactly true: under volatility clustering, which Chapter 38 models when it forecasts volatility, the size of tomorrow's move is partly predictable even though its direction is not. The Markov assumption is nevertheless close enough to be the standard starting point.

The model adds one assumption this chapter relies on throughout: changes over non-overlapping periods are independent, so their variances add. The Markov property alone does not deliver this: the mean-reverting interest rates of Chapter 31 are Markov, yet their changes are negatively correlated, and their variance levels off instead of growing linearly.

Example 14.1 Why variances, not standard deviations, add

Suppose a stock at \$40 has a price change with standard deviation \$10 over one year. What is the uncertainty over two years?

The change over two years is the change in year one plus the change in year two. These are assumed independent, so their *variances* add:

$$\text{Var}(2\text{ years}) = 100 + 100 = 200, \qquad \text{standard deviation} = \sqrt{200} = \$14.14$$

Uncertainty, measured by standard deviation, grows with $\sqrt{T}$, not with T; it is the variance that grows with T. Over six months the standard deviation is $10\sqrt{0.5} = \$7.07$; over three months, $10\sqrt{0.25} = \$5.00$.

This fact is why Section 11.1 found option value growing roughly as $\sqrt{T}$. It is why volatility is quoted per annum and rescaled by $\sqrt{\Delta t}$. And it is why the binomial parameters of Section 13.8 were $u = e^{\sigma\sqrt{\Delta t}}$ rather than $e^{\sigma\Delta t}$. The square root was not a modeling choice but a consequence of independence.

14.2 WIENER PROCESSES

The building block is a process W with two properties. Over a short interval Δt,

$$\Delta W = \varepsilon\sqrt{\Delta t}, \qquad \varepsilon \sim N(0,1) \tag{14.1}$$

and the values of ΔW for any two non-overlapping intervals are independent. A process with these properties is a **Wiener process**, also called **Brownian motion**.

Two consequences follow immediately. The mean of ΔW is zero and its variance is Δt, so the variance per unit time is exactly 1, which is what makes W a natural unit of randomness. And over a long interval T, made of $T/\Delta t$ independent steps,

$$W(T) - W(0) \sim N(0, T)$$

by the same variance-addition argument as Example 14.1.

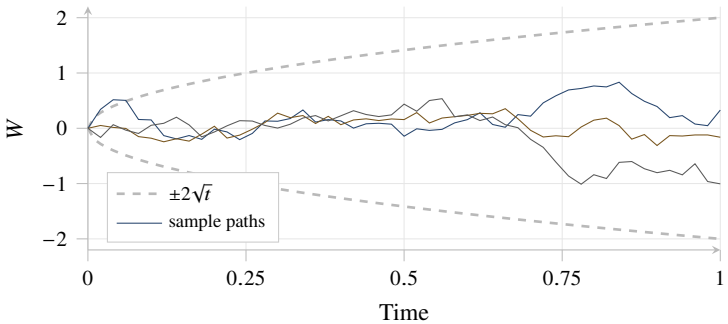


Figure 14.1: Three illustrative simulated Wiener paths and the $\pm 2\sqrt{t}$ envelope.

Two features of the paths in Figure 14.1 recur throughout Part III.

The paths are continuous but extremely rough. As Δt shrinks, the standard deviation of ΔW shrinks only as $\sqrt{\Delta t}$, so that of $\Delta W/\Delta t$ is $1/\sqrt{\Delta t}$ and grows without bound. A Wiener path has, with probability one, no derivative anywhere. This is why the calculus of Section 14.7 has to be different from the ordinary kind: one cannot simply write dW/dt.

And the envelope widens as $\sqrt{t}$, not linearly. The typical distance from zero grows as time passes, but at a decreasing rate.

14.3 GENERALIZED WIENER PROCESSES

A Wiener process has zero **drift**, the expected change per unit time, and a unit **variance rate**, the variance per unit time. Both can be generalized. A variable x follows a **generalized Wiener process** if

$$dx = a\,dt + b\,dW \tag{14.2}$$

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with a and b constants. The $a\,dt$ term is a deterministic trend (x drifts at rate a per unit time), and the $b\,dW$ term, the **diffusion**, adds noise with variance rate b^2.

Over an interval T,

$$x(T) - x(0) \sim N\left(aT,\ b^2T\right)$$

The mean grows linearly in T and the standard deviation as $\sqrt{T}$, so the trend eventually dominates the noise, but only eventually, and the crossover can be a long way off, as Example 14.2 shows.

Example 14.2 A cash position

A company's cash balance, in millions, follows a generalized Wiener process with drift 20 per year and variance rate 900 per year. It currently holds \$50 million.

After one year the balance is distributed $N(50 + 20,\ 900)$, that is, mean \$70 million with standard deviation \$30 million. After six months it is $N(60,\ 450)$: mean \$60 million, standard deviation $\sqrt{450}$ = \$21.2 million.

Writing $N(x)$ for the standard normal cumulative distribution function, the probability that the balance is negative at the end of one year is

$$\Pr(x < 0) = N\left(\frac{0 - 70}{30}\right) = N(-2.333) = 0.98\%$$

and at the end of six months, $N(-60/21.2) = N(-2.83) = 0.23\%$. The balance is expected to *grow*, yet the probability is *higher* at one year than at six months, because uncertainty accumulates faster than the drift does over short horizons.

Running out of cash, though, means touching zero at any time, not being below it on a given date. That probability is larger, 2.7% within the year (by the reflection argument Chapter 34 uses for barrier options), and it never falls: over an unlimited horizon it is $e^{-2\times20\times50/900} = 10.8\%$. This is the same structure as the funding failures of Chapter 8: a position with a positive expected trend can still fail on the way, and the distribution on any one date understates how often.

An **Itô process** generalizes (14.2) once more by letting the coefficients depend on the current value and on time:

$$dx = a(x,t)\,dt + b(x,t)\,dW \tag{14.3}$$

This is the form most of the rest of the book uses. It is shorthand for an integral equation,

$$x(T) - x(0) = \int_0^T a(x,t)\,dt + \int_0^T b(x,t)\,dW$$

and the second integral is not an ordinary one, because W has no derivative to integrate against. It is an **Itô integral**, the limit of the sums

$$\sum_{i=0}^{n-1} b\big(x(t_i), t_i\big)\,\Delta W_i, \quad \Delta W_i = W(t_{i+1}) - W(t_i), \quad 0 = t_0 < \cdots < t_n = T,$$

as the longest interval shrinks to zero, in which b is evaluated at the *start* of each interval, before the increment ΔW_i is known. In finance this is not a convention but a fact about trading: a hedge ratio is chosen before the price move it is exposed to, so a trading gain is a sum of exactly this kind. The choice makes the integral a fair game (a martingale, the term Section 13.6 defined), with expected value zero whatever b does, and it is the source of everything that separates Itô calculus from the ordinary kind, beginning with the extra term in Section 14.7. Evaluating b at the midpoint of each interval instead gives the Stratonovich integral, which obeys the ordinary chain rule but is not a fair game and does not describe what a trader can earn. The derivation in [TN14] works through the **quadratic variation** of W, the limit of the sum of its squared increments, which over $[0, T]$ is exactly T; Section 14.7 shows why.

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14.4 THE PROCESS FOR A STOCK PRICE

What process should a stock price follow? A generalized Wiener process is the obvious candidate, and it is wrong for two reasons that determine the model that replaces it.

Constant absolute drift is not how returns work. A \$10 stock and a \$1,000 stock do not have the same expected dollar gain. Investors require a *percentage* return, so the drift should be proportional to the price.

Constant absolute volatility is not how risk works either. A \$1 move means something very different at \$10 than at \$1,000. And a constant b permits the price to go negative, which for a limited-liability share it cannot.

Both objections are met by making the coefficients proportional to S:

$$\boxed{dS = \mu S\,dt + \sigma S\,dW} \tag{14.4}$$

This is **geometric Brownian motion**, and it is the model underlying everything from Chapter 15 onward. Dividing through, the *proportional* return over a short Δt is close to normal:

$$\frac{dS}{S} = \mu\,dt + \sigma\,dW, \qquad \frac{\Delta S}{S} \sim N\left(\mu\,\Delta t,\ \sigma^2 \Delta t\right).$$

Here μ is the expected return per year and σ is the volatility, the same σ that Section 13.8 fed into the binomial tree.

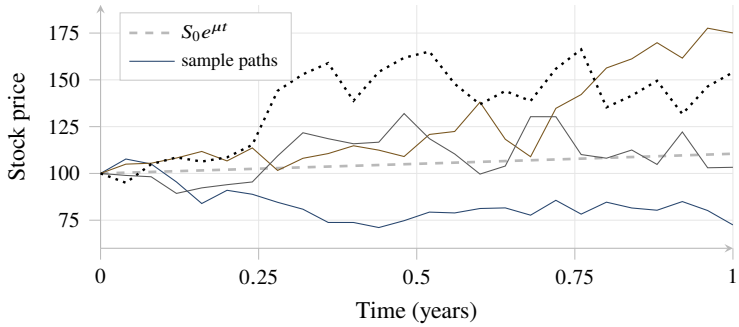


Figure 14.2: Four illustrative simulated paths. $S_0 = 100$, $\mu = 10\%$, $\sigma = 40\%$.

Figure 14.2 shows four paths from this process. Note that none of them tracks the dashed line $S_0 e^{\mu t}$ at all closely, and that half of them finish below it despite μ being positive. Section 14.9 explains why that is exactly what should be expected.

Example 14.3 Reading the process over a short interval

A stock trades at \$50 with $\mu = 16\%$ and $\sigma = 30\%$. Over one week, $\Delta t = 1/52 = 0.01923$:

$$\frac{\Delta S}{S} \sim N\left(0.16 \times 0.01923,\ 0.30^2 \times 0.01923\right) = N(0.003077,\ 0.001731)$$

so the weekly return has mean 0.31% and standard deviation $\sqrt{0.001731} = 4.16\%$. In dollar terms,

$$\Delta S \sim N(0.154,\ 2.08^2)$$

The noise term is about thirteen and a half times the drift term over one week, and since the drift scales with Δt while the noise scales with $\sqrt{\Delta t}$, the ratio gets worse as the interval shortens: over one day it is about thirty to one.

This is why estimating μ from historical returns is hopeless in a way that estimating σ is not: to measure a drift buried under thirty times its own size in noise, one needs decades of data, over which the drift will not have stayed constant.

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The discrete-time version

Equation (14.4) is a statement about infinitesimals. To simulate it, or to connect it to Chapter 13's tree, it must be discretized, and there are two ways of doing so that are not equivalent.

The direct reading of (14.4) gives

$$\Delta S \approx \mu S\,\Delta t + \sigma S\,\varepsilon\sqrt{\Delta t} \tag{14.5}$$

which is accurate only for small Δt and can produce a negative price if ε is sufficiently negative. The alternative uses the solution of the equation rather than the equation itself, which Section 14.9 will derive:

$$S_{t+\Delta t} = S_t \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)\Delta t + \sigma\varepsilon\sqrt{\Delta t}\right] \tag{14.6}$$

This is *exact* for any Δt, and it cannot go negative. Figure 14.2 was generated with (14.6).

The comparison is worth keeping in mind because the same choice arises whenever a process is discretized, and Chapter 20 returns to it. Note also that (14.6) is recognizably Chapter 13's tree with the two-point distribution of ε replaced by a normal one: $u = e^{\sigma\sqrt{\Delta t}}$ was the same expression with $\varepsilon = \pm 1$ and the drift moved out of the step and into p.

Where the model came from. Louis Bachelier's thesis of 1900 used the arithmetic process of Section 14.3, which this section has just rejected for stocks. Norbert Wiener gave the process its rigorous construction in 1923. Kiyoshi Itô defined the integral of Section 14.3 in 1944 and set out the calculus built on it, including the lemma of Section 14.7, in 1951. Section 15.12 traces the economics, from Bachelier to Black, Scholes, and Merton.

14.5 EXPECTED RETURN

The parameter μ was introduced as "the expected return," and that phrase turns out to name two different numbers. The distinction causes more confusion than any other point in this chapter, and it is not confined to theory.

From (14.4), the expected *proportional* return over a short interval satisfies

$$\mathbb{E}\left[\frac{\Delta S}{S}\right] \approx \mu\,\Delta t$$

so μ is the expected return in the sense of an arithmetic average of short-period returns. But from (14.10), which Section 14.9 derives, the expected *continuously compounded* return over $[0, T]$ is

$$\mathbb{E}\left[\frac{1}{T}\ln\frac{S_T}{S_0}\right] = \mu - \frac{\sigma^2}{2}$$

which is the expected rate earned by an investor who holds the asset throughout. Example 14.4 shows the gap in five years of returns.

Example 14.4 Arithmetic and geometric averages

An asset returns +30%, −10%, +20%, −5%, and +15% over five years. The arithmetic average is

$$\frac{30 - 10 + 20 - 5 + 15}{5} = 10.0\%$$

But an investor who held throughout finished with

$$1.30 \times 0.90 \times 1.20 \times 0.95 \times 1.15 = 1.53387$$

times their money, an annual compounded return of $1.53387^{1/5} - 1 = 8.93\%$.

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In log returns the five years average 8.56% ($e^{0.0856} - 1 = 8.93\%$), 1.44 percentage points below the arithmetic average, and half the sample variance of the five simple returns is $0.0288/2 = 1.44$ percentage points, a match to the second decimal that in general holds only approximately. This is the $\sigma^2/2$ that separates (14.13) from (14.14), seen in five numbers.

The arithmetic average is not wrong; it answers a different question. It is the right input for an expected one-period return, and therefore the right thing to compare with μ. The compounded figure is the right answer to "what did an investor make," and it corresponds to $\mu - \sigma^2/2$.

The practical consequence is a reporting convention that repays skepticism. A track record summarized by the arithmetic average of its annual returns, common in strategy marketing and backtests, exceeds what an investor who held throughout received, and the excess grows with volatility. The standardized figure a US mutual fund must publish is not of this kind: the SEC's Form N-1A defines its average annual total return as the compounded rate.

14.6 VOLATILITY

As Section 13.8 defined it, σ is the annualized standard deviation of the continuously compounded return. Over a period Δt it is $\sigma\sqrt{\Delta t}$, by the variance-addition property of Section 14.1.

One rule of thumb is used constantly on trading desks. With about 252 trading days in a year and $\sqrt{252} \approx 15.9$, the daily standard deviation is approximately

$$\sigma_{\text{daily}} \approx \frac{\sigma}{16}$$

so a stock with 32% annual volatility has a daily standard deviation of about 2%, and a 6% move is a three-standard-deviation event. It converts an abstract parameter into something a trader can check against the screen.

How σ is estimated (from historical returns, or by inverting an option price) is taken up in Chapter 15, where it is needed for the first time.

14.7 ITÔ'S LEMMA

Assume that a derivative's value depends only on the current stock price and time, so that it can be written $f(S, t)$. That is an assumption, not a general truth: it holds for standard calls and puts, and fails for contracts whose payoff depends on the path, such as Chapter 34's Asian and lookback options, whose value depends on where the stock has been and not only on where it is now. If S follows (14.4), what process does $f(S, t)$ follow? Ordinary calculus gives the wrong answer, and the reason it does is the single technical fact this chapter exists to deliver.

Consider a Taylor expansion of f:

$$\Delta f = \frac{\partial f}{\partial S}\Delta S + \frac{\partial f}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\Delta S^2 + \cdots$$

In ordinary calculus the ΔS^2 term is discarded, because ΔS is proportional to Δt and Δt^2 is negligible. Here it is not. From (14.4), $\Delta S \approx \mu S\,\Delta t + \sigma S\varepsilon\sqrt{\Delta t}$, so

$$\Delta S^2 = \sigma^2 S^2 \varepsilon^2\,\Delta t + \text{terms of order } \Delta t^{3/2} \text{ and smaller}$$

whose leading term is of order Δt, *not* Δt^2. It survives. A single $\varepsilon^2\Delta t$ is still random: its mean is Δt and its standard deviation $\sqrt{2}\,\Delta t$, of the same size. But over an interval T, made of $T/\Delta t$ steps, the squared increments add to a total with mean T and variance $2T\Delta t$, which vanishes as the steps shrink. The sum, the quadratic variation of Section 14.3, becomes non-random and equal to T, as Figure 14.3 shows on one path. In the notation that results,

$$dW^2 = dt \tag{14.7}$$

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a rule about sums over many steps, not about any single increment. [TN14], reproduced in Appendix A, makes this precise.

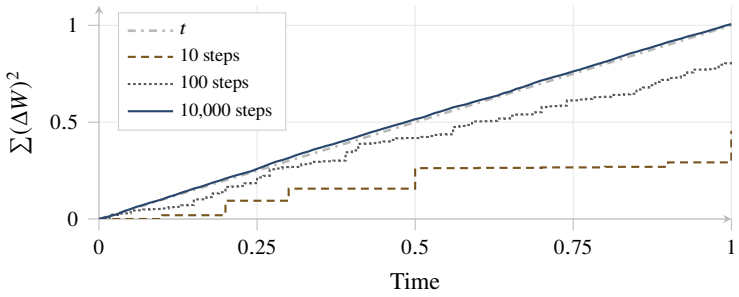


Figure 14.3: Running sum of squared increments along one simulated Wiener path, sampled on grids of 10, 100 and 10,000 steps.

Substituting dS and $dS^2 = \sigma^2 S^2\,dt$ into the expansion gives **Itô's lemma**:

$$\boxed{df = \left(\frac{\partial f}{\partial S}\mu S + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial S^2}\sigma^2 S^2\right)dt + \frac{\partial f}{\partial S}\sigma S\,dW} \tag{14.8}$$

Everything except the third term in the drift is what ordinary calculus would give. That third term is the entire difference, and it is the reason a derivative's expected return is not simply the chain rule applied to the stock's. It is proportional to the second derivative, the convexity, which is why it will reappear in Chapter 18 as gamma and in Chapter 30 as the source of convexity adjustments.

Example 14.5 Itô's lemma applied to a forward price

The forward price of a stock that pays no dividend, for delivery at T is $F = Se^{r(T-t)}$, from Section 5.2. What process does F follow?

The partial derivatives are

$$\frac{\partial F}{\partial S} = e^{r(T-t)}, \qquad \frac{\partial F}{\partial t} = -rSe^{r(T-t)}, \qquad \frac{\partial^2 F}{\partial S^2} = 0$$

Substituting into (14.8), and noting that the second-derivative term vanishes:

$$dF = \left(e^{r(T-t)}\mu S - rSe^{r(T-t)}\right)dt + e^{r(T-t)}\sigma S\,dW = (\mu - r)F\,dt + \sigma F\,dW$$

So the forward price also follows geometric Brownian motion, with *the same volatility* σ as the spot but a drift of $\mu - r$ rather than μ.

The equality of volatilities is why a volatility quoted on the spot can be used to price an option on the forward, which Chapter 16 relies on throughout. And the drift being the *excess* return $\mu - r$ is the continuous-time statement of Section 13.10's observation that a futures price has zero growth in the risk-neutral world: setting $\mu = r$, as risk-neutral valuation does, makes the forward's drift exactly zero.

Common applications

Table 14.1 collects the cases used elsewhere in the book. Each is (14.8) with the relevant derivatives substituted, and each is worth being able to reproduce.

14.8 WHY THE LEMMA DELIVERS A HEDGE

The last row of Table 14.1 is the bridge between Chapter 13 and Chapter 15.

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Table 14.1: Processes followed by functions of a stock price obeying (14.4).

Function	Process	Where used
$\ln S$	$\left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma\, dW$	The lognormal property, Section 14.9
S^n	$S^n\left[\left(n\mu + \frac{n(n-1)}{2}\sigma^2\right) dt + n\sigma\, dW\right]$	Power payoffs; $n = 2$ in Practice Question 5
$Se^{r(T-t)}$	$(\mu - r)F\, dt + \sigma F\, dW$	Forward prices, Example 14.5
general $f(S,t)$	drift as in (14.8), diffusion $\dfrac{\partial f}{\partial S}\sigma S\, dW$	Every derivative, Section 14.8

For *any* derivative $f(S,t)$, Itô's lemma gives

$$df = (\cdots)\, dt + \frac{\partial f}{\partial S}\, \sigma S\, dW \tag{14.9}$$

Whatever the drift term turns out to be, the *random* part of the derivative's movement is the stock's own shock dW, scaled by $\sigma S\, \partial f/\partial S$. And the random part of the stock's movement, from (14.4), is $\sigma S\, dW$: the same dW.

Consider a portfolio that is short one derivative and long $\partial f/\partial S$ shares. Its change over an infinitesimal time interval is

$$-df + \frac{\partial f}{\partial S}\, dS = -(\cdots)\, dt - \frac{\partial f}{\partial S}\sigma S\, dW + \frac{\partial f}{\partial S}\, (\mu S\, dt + \sigma S\, dW)$$

and the two dW terms cancel exactly. The portfolio has no random component at all. It is riskless, so it must earn r, and imposing that condition is precisely what Chapter 15 does to obtain the Black–Scholes–Merton equation.

Three observations follow, and they connect this chapter back to the last one.

It is Chapter 13's argument in continuous time. Section 13.1 found the Δ that made a portfolio worth the same in both states. Here $\partial f/\partial S$ makes a portfolio's change contain no dW. The discrete hedge ratio

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}$$

is the difference quotient whose limit is $\partial f/\partial S$.

μ cancels too. The derivative's drift, the $(\cdots)$ above, contains the term $\mu S\, \partial f/\partial S$; the short derivative subtracts it and the shares add it back, so μ drops out of the portfolio's change along with dW, leaving only the time and convexity terms for Chapter 15 to set equal to the riskless return. This is the continuous-time version of Chapter 13's observation that the real-world probability never entered, and it is why an option can be priced without forecasting the stock.

The hedge is instantaneous, not permanent. The cancellation holds over a single dt. Since $\partial f/\partial S$ changes as S and t change, the position must be adjusted continuously: the dynamic replication of Section 13.5, now requiring infinitely frequent trading. That requirement is harmless in the model and expensive in reality, which is the subject of Chapter 18 and the reason the assumptions of Section 14.12 matter.

14.9 THE LOGNORMAL PROPERTY

The most useful application of Itô's lemma is to $G = \ln S$. Here

$$\frac{\partial G}{\partial S} = \frac{1}{S}, \qquad \frac{\partial^2 G}{\partial S^2} = -\frac{1}{S^2}, \qquad \frac{\partial G}{\partial t} = 0$$

and (14.8) gives

$$d\ln S = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma\, dW \tag{14.10}$$

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The right-hand side has constant coefficients, so $\ln S$ follows a *generalized* Wiener process, and Section 14.3 already told us the answer for those:

$$\ln S_T \sim N\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T,\ \ \sigma^2 T\right) \tag{14.11}$$

Because its logarithm is normal, S_T is lognormal. This is the result Chapter 13 anticipated by a different route: Section 13.9 showed that a product of many small independent multiplicative shocks becomes lognormal by the central limit theorem. Equation (14.11) is that conclusion in continuous time, and the agreement is not a coincidence: the binomial model was built to converge to this one.

Moments of the lognormal

Two results are used constantly from Chapter 15 onward. If $\ln X \sim N(m, s^2)$, then

$$\mathbb{E}[X] = e^{m+s^2/2}, \qquad \mathrm{Var}(X) = e^{2m+s^2}\left(e^{s^2} - 1\right) \tag{14.12}$$

Applying the first to (14.11), with $m = \ln S_0 + (\mu - \sigma^2/2)T$ and $s^2 = \sigma^2 T$:

$$\mathbb{E}[S_T] = \exp\left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T + \frac{\sigma^2 T}{2}\right) = S_0 e^{\mu T}$$

The two rates are consistent: μ is the expected growth rate of the price, and Itô's lemma makes the logarithm, and with it the median, grow $\sigma^2/2$ more slowly.

Where the $\sigma^2/2$ comes from, and why it matters

The term $-\sigma^2/2$ in (14.10) is the Itô correction term of (14.8), and it is the most consequential piece of notation in this chapter.

From (14.11), two different quantities both have a claim to be called "the" growth rate:

$$\mathbb{E}[S_T] = S_0 e^{\mu T} \tag{14.13}$$

$$\mathrm{median}(S_T) = S_0 e^{(\mu - \sigma^2/2)T} \tag{14.14}$$

The expected value grows at μ. The median (and the continuously compounded return actually realized on a typical path) grows only at $\mu - \sigma^2/2$. The gap is entirely due to volatility, and it is sometimes called **volatility drag**.

Example 14.6 Expected value against typical outcome

A stock trades at \$100 with $\mu = 10\%$ and $\sigma = 40\%$. After one year:

$$\mathbb{E}[S_T] = 100e^{0.10} = \$110.52, \qquad \text{median} = 100e^{0.10-0.08} = 100e^{0.02} = \$102.02$$

The expected price is \$110.52 and yet half of all outcomes fall below \$102.02. Worse, ask how often the expected value is reached at all. Since $\ln S_T \sim N(\ln 100 + 0.02,\ 0.40^2)$,

$$\Pr(S_T > 110.52) = 1 - N\left(\frac{\ln 1.1052 - 0.02}{0.40}\right) = 1 - N(0.20) = 42.1\%$$

The stock finishes below its own expected value 58% of the time. Nothing is wrong with the arithmetic: the mean is dragged up by a long right tail that occurs rarely.

A 95% interval for the terminal price makes the dispersion concrete. With probability 0.95,

$$\ln 100 + 0.02 - 1.96 \times 0.40 \le \ln S_T \le \ln 100 + 0.02 + 1.96 \times 0.40,$$

that is, $\$46.58 \le S_T \le \$223.45$ at 40% volatility over a single year.

Two common errors follow directly from ignoring the distinction, besides the fund-reporting one of Example 14.4. Quoting an "expected return" of 10% invites investors to expect \$110.52,

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when the typical outcome is \$102.02. And a structured product promising participation in an index's *average* growth is offering something systematically better than participation in its *realized* growth, a difference of $\sigma^2/2$ per year, which at 40% volatility is eight percentage points.

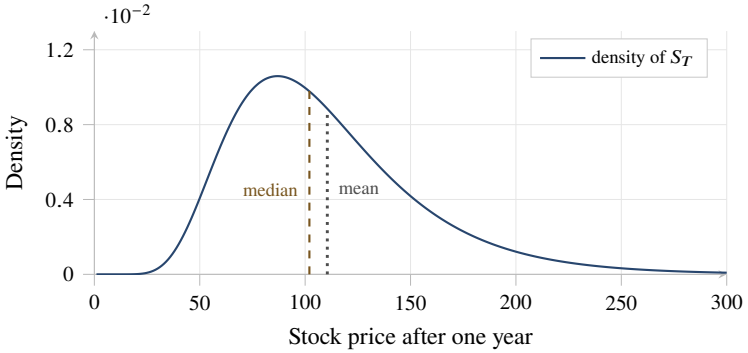


Figure 14.4: Density of S_T. $S_0 = 100$, $\mu = 10\%$, $\sigma = 40\%$, $T = 1$.

Figure 14.4 shows why. The distribution is bounded below by zero and has a long right tail; the mean sits to the right of the median, dragged there by outcomes that are large but rare. The mode is lower still, at $S_0 e^{(\mu - 3\sigma^2/2)T} = \$86.94$.

Which rate does an option price use? Neither: Section 14.10 replaces μ by r, and the $-\sigma^2/2$, a property of the lognormal distribution rather than of the drift, survives.

Case Study 14.1 Leveraged exchange-traded funds

A widely sold class of funds promises a fixed multiple (usually two or three times) of the *daily* return of an index. Investors often assume this delivers the same multiple over longer horizons. It does not, and Itô's lemma says exactly by how much it fails.

If the index follows (14.4), a fund whose value V delivers k times its instantaneous return has $dV/V = k\,dS/S = k\mu\,dt + k\sigma\,dW$, a geometric Brownian motion with drift $k\mu$ and volatility $k\sigma$, before the financing cost of the borrowed $k - 1$ units of exposure, which lowers the drift by a further $(k-1)r$. From (14.14), its median grows at

$$k\mu - \frac{(k\sigma)^2}{2} = k\mu - \frac{k^2\sigma^2}{2}$$

while k times the index's own median growth would be $k\left(\mu - \sigma^2/2\right)$. The shortfall is

$$\frac{k(k-1)}{2}\,\sigma^2 \text{ per year}$$

which for $k = 2$ is σ^2 and for $k = 3$ is $3\sigma^2$.

Numbers make the size clear. Take an index with $\mu = 8\%$ and $\sigma = 25\%$. Over one year the index's median return is $e^{0.08 - 0.03125} - 1 = 5.0\%$, so an investor might expect roughly 10% from a two-times fund. The fund's median return is

$$e^{0.16 - 0.125} - 1 = 3.6\%$$

At $\sigma = 40\%$ the arithmetic turns brutal: the index's median return is $e^{0.08 - 0.08} - 1 = 0\%$, and the two-times fund's median is $e^{0.16 - 0.32} - 1 = -14.8\%$. In the median case the index goes nowhere and the leveraged fund loses fifteen percent, before financing, fees, and the cost of daily rebalancing.

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> Nothing here is a defect in the product, which does precisely what its prospectus says: it delivers k times the return *each day*. The gap is a mathematical consequence of compounding a scaled-up volatility, and it is the same $\sigma^2/2$ that separates the mean from the median in Example 14.6. These funds are sound instruments for a position held for days and poor ones for a position held for years, and the boundary between the two is set by σ.

14.10 THE RISK-NEUTRAL PROCESS

Chapter 13 established that a derivative may be valued as though every asset earned the risk-free rate. In the language of this chapter that statement becomes a single substitution: under the risk-neutral measure Q (Section 13.6), with W^Q a Wiener process under Q,

$$dS = rS\,dt + \sigma S\,dW^Q \tag{14.15}$$

and therefore, repeating Section 14.9 with μ replaced by r,

$$\ln S_T \sim N\left(\ln S_0 + \left(r - \frac{\sigma^2}{2}\right)T,\ \ \sigma^2 T\right), \qquad \mathbb{E}^Q[S_T] = S_0 e^{rT} \tag{14.16}$$

Equation (14.16) is the entire input to Chapter 15. Everything the Black–Scholes–Merton formula knows about the stock is contained in it.

Only the drift changes. The volatility σ is identical under both measures, as Section 13.8 found for the tree. The formal statement is Girsanov's theorem (a change to an equivalent measure adds a drift to a Wiener process and changes nothing else), which [TN14] sketches; the intuition is that changing the measure re-weights which paths are likely but cannot change how rough they are.

The correction term survives. Substituting r for μ does not remove the $-\sigma^2/2$, because that term came from Itô's lemma rather than from the drift. This is why the Black–Scholes–Merton formula contains $r - \sigma^2/2$; the correction is the Itô term of the logarithm, not a property of the expected return.

The connection back to Chapter 13 is exact. There, p made the discounted stock price a martingale (Section 13.6), the condition that the stock's expected growth be r. Here, $\mathbb{E}^Q[S_T] = S_0 e^{rT}$ is the same condition, written for a continuum of states.

14.11 TWO SOURCES OF UNCERTAINTY

Many instruments depend on more than one uncertain quantity: a quanto (a foreign asset's payoff converted at an exchange rate fixed in advance, Chapter 16) on both an asset and an exchange rate, a spread option (an option on the difference between two prices) on two prices. The apparatus needs one addition.

Let W_1 and W_2 be Wiener processes whose increments have correlation ρ, so that, by the argument behind (14.7),

$$dW_1\,dW_2 = \rho\,dt \tag{14.17}$$

alongside $dW_1^2 = dW_2^2 = dt$ from (14.7). For a function $f(x, y, t)$ of two Itô processes $dx = a_1 dt + b_1 dW_1$ and $dy = a_2 dt + b_2 dW_2$, Itô's lemma becomes

$$\begin{aligned} df = {} & \left(\frac{\partial f}{\partial x}a_1 + \frac{\partial f}{\partial y}a_2 + \frac{\partial f}{\partial t} + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}b_1^2 + \frac{1}{2}\frac{\partial^2 f}{\partial y^2}b_2^2 + \frac{\partial^2 f}{\partial x\,\partial y}b_1 b_2 \rho\right)dt \\ & + \frac{\partial f}{\partial x}b_1\,dW_1 + \frac{\partial f}{\partial y}b_2\,dW_2 \end{aligned} \tag{14.18}$$

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The only new term, the last in the drift, exists for the same reason as the two second-derivative terms: the cross-product $\Delta x\,\Delta y$ is of order Δt rather than Δt^2. It is also the source of the correlation term in Chapter 30's quanto adjustment. Example 14.7 applies it to a price ratio.

Example 14.7 The volatility of a ratio

Two assets follow geometric Brownian motions with expected returns μ_1 and μ_2, volatilities σ_1 and σ_2, and correlation ρ. What process does the ratio $G = S_1/S_2$ follow?

Applying (14.18) and collecting terms gives

$$\frac{dG}{G} = \left(\mu_1 - \mu_2 + \sigma_2^2 - \rho\sigma_1\sigma_2\right) dt + \sigma_1\, dW_1 - \sigma_2\, dW_2$$

so the ratio is itself a geometric Brownian motion, with volatility

$$\sigma_G = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \tag{14.19}$$

Put numbers on it. With $\sigma_1 = 20\%$, $\sigma_2 = 12\%$, and $\rho = 0.4$:

$$\sigma_G = \sqrt{0.04 + 0.0144 - 2 \times 0.4 \times 0.20 \times 0.12} = \sqrt{0.0352} = 18.8\%$$

The ratio is *less* volatile than S_1 alone, because the two assets partly move together and the common movement cancels in the ratio. Reverse the correlation to $\rho = -0.4$ and $\sigma_G = \sqrt{0.0544 + 0.0192} = \sqrt{0.0736} = 27.1\%$, more volatile than either input.

Equation (14.19) is used more often than almost any other result in this chapter:

- It gives the volatility of a cross exchange rate from two rates against a common currency.
- It gives the volatility of a stock measured in a foreign currency, which Chapter 16 needs for composite options (options on a foreign stock whose payoff is converted at the expiration exchange rate). A product S_1S_2 is the ratio of S_1 to $1/S_2$, whose correlation with S_1 is $-\rho$, so the minus sign in (14.19) becomes a plus.

It is the formula for the standard deviation of a difference, which is what a ratio becomes after taking logarithms. That is the quickest way to remember it.

14.12 WHAT THE MODEL ASSUMES

Geometric Brownian motion is a modeling choice, not a fact. Section 15.1 collects the assumptions of the pricing model built on it and says where each fails; three ways real prices depart from it are worth having here. Volatility is not constant but clusters (calm follows calm and violent follows violent), with consequences Chapter 19 documents in the volatility surface. Prices are not continuous: the negative oil settlement of Section 5.9, the LME nickel suspension of Chapter 2 and any earnings announcement produce moves no continuous path can accommodate, and a model with jumps assigns much more probability to extreme outcomes than (14.11) does. And returns are independent across time in direction but not in magnitude: clustering violates the Markov property of Section 14.1 for the price alone, though not for price and volatility jointly, and Section 14.13 introduces the standard process whose increments are not independent. Of the two parameters, σ must be estimated (Chapter 15 takes up how), while Example 14.3 showed that μ effectively cannot be, which does not matter for option pricing and matters a great deal for everything else.

14.13 FRACTIONAL BROWNIAN MOTION

Rough volatility, the finding that the path of an asset's volatility is far more jagged than a Wiener path, together with the models built on that finding, appears in Sections 21.8 and 23.6 and in [TN21], so the process behind it is introduced here. It is also the standard process for which Section 14.1's variance rule fails.

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Definition. Fractional Brownian motion B_t^H is a continuous process, jointly normal at any set of dates, with $B_0^H = 0$, mean zero, and covariance

$$\mathrm{Cov}\left[B_s^H, B_t^H\right] = \frac{1}{2}\left(s^{2H} + t^{2H} - |t-s|^{2H}\right), \qquad 0 < H < 1 \tag{14.20}$$

Its increments are **stationary**, as a Wiener process's are: the variance of a change depends only on the length of the interval,

$$\mathrm{Var}\left[B_t^H - B_s^H\right] = |t-s|^{2H} \tag{14.21}$$

The exponent H is the **Hurst parameter**, after the hydrologist who first measured it in the flood records of the Nile. At $H = 1/2$ the covariance (14.20) is $\min(s,t)$, which is the Wiener process of Section 14.2: the variance over T is T and increments over disjoint intervals are independent.

The increments are correlated. Take two consecutive years. The change over the second year, $B_2^H - B_1^H$, has by stationarity the same variance as the change over the first, and the two-year variance is the sum of the two plus twice their covariance. With (14.21) on both sides, $2^{2H} = 1 + 1 + 2\rho$, so the correlation between successive equal increments is

$$\rho = 2^{2H-1} - 1 \tag{14.22}$$

which is -0.43 at $H = 0.1$, zero at $1/2$, and $+0.74$ at $H = 0.9$. Below one-half the process is **anti-persistent**: each move partly undoes the last. Above one-half it is **persistent**, with a long memory: a rise makes a further rise more likely. Section 14.1's argument that variances add needed exactly this cross term to vanish, and it vanishes only at one-half.

Scaling and roughness. Equation (14.21) says the process is **self-similar**: stretching time by a factor c scales the path's distribution by c^H, so an increment over a short interval Δt has standard deviation Δt^H. For the Wiener process that is $\sqrt{\Delta t}$, the scaling Section 14.2 built on. For $H < 1/2$, Δt^H is larger than $\sqrt{\Delta t}$ for small Δt, so the path moves more over short intervals and is *rougher*; for $H > 1/2$ it is smoother. Figure 14.5 shows the three cases built from the same random draws: the rough path reverses constantly, the persistent one trends, and the Wiener path sits between them.

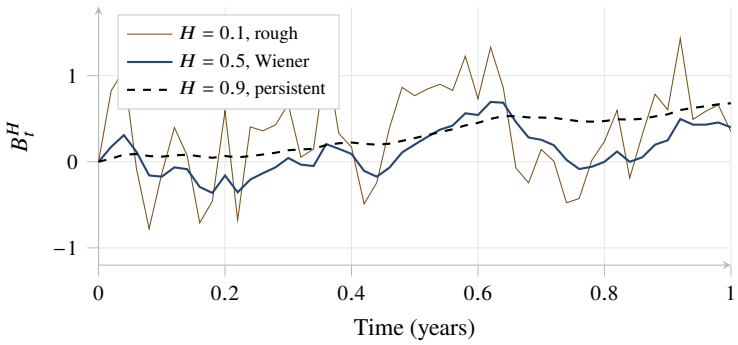


Figure 14.5: Fractional Brownian motion at $H = 0.1$, 0.5 and 0.9, built from the same random draws; each has unit variance at one year.

The roughness matters for the calculus of this chapter. Section 14.7 rested on $dW^2 = dt$: the statement that the squared increments of a Wiener path over $[0,T]$ add up to T. For fractional Brownian motion the n squared increments over a grid of step T/n add up to about

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$n\,(T/n)^{2H} = T^{2H}\,n^{1-2H}$, which grows without bound for $H < 1/2$ and shrinks to zero for $H > 1/2$. There is no $dB^2 = dt$, and Itô's lemma in the form of Section 14.7 does not apply. Nor is the process Markov: by (14.20) the next increment is correlated with every past increment, so the current value does not summarize the path and forecasting needs the whole history.

Example 14.8 Variances that do not add

Take $H = 0.1$, the value Chapter 21 estimates for log-volatility, and unit variance over one year.

Two years. By (14.21) the two-year variance is $2^{0.2} = 1.15$, against 2 for a Wiener process, and the two-year standard deviation is $2^{0.1} = 1.07$ against $\sqrt{2} = 1.41$. Doubling the horizon adds 7% to the uncertainty, not 41%, because the second year's move is correlated at −0.43 with the first and partly undoes it.

Annualizing daily data. Section 14.6 turned a daily standard deviation into an annual one by multiplying by $\sqrt{252}$. For a process with Hurst parameter H the factor is 252^H instead: $252^{0.1} = 1.74$ at $H = 0.1$, against $252^{0.5} = 15.9$. Applying the square-root rule to such a series would overstate the one-year standard deviation by $252^{0.4} = 9.1$ times. The same arithmetic at $H = 0.9$ understates it by the same factor. The square-root-of-time rule is the $H = 1/2$ case of T^H, and it is only as good as the independence it assumes.

Built from a Wiener process. One standard construction writes the process as a weighted sum of past Wiener increments,

$$B_t^H = \sqrt{2H}\int_0^t (t-s)^{H-1/2}\,dW_s \tag{14.23}$$

The kernel $(t-s)^{H-1/2}$ weights each past shock by its age. At $H = 1/2$ it is one, and the integral is W_t. Below one-half the exponent is negative and the weight on the most recent shocks rises without bound as $s \to t$, which is where the roughness comes from, while the weight on old shocks decays as a power of their age rather than dying off exponentially, which is where the memory comes from. The $\sqrt{2H}$ makes the variance at each date exactly t^{2H}, but this version, started at zero, has stationary increments only approximately; it is the Riemann–Liouville form, a close relative. Mandelbrot and Van Ness's two-sided integral has the exact covariance (14.20) and stationary increments, and nothing in this book turns on the difference. Section 21.8 uses this kernel, with the integral inside an exponential and a volatility-of-volatility in front.

What it can and cannot model. As a model for a traded price it fails, and for a reason the correlation (14.22) makes plain: a process whose next move is predictable from its last can be traded on. With $H > 1/2$ a rise predicts a rise, with $H < 1/2$ a rise predicts a fall, and in either case a strategy that trades on the prediction at ever shorter intervals earns a profit with a risk that shrinks toward zero. Trading gains cannot be defined as Itô integrals against it (in the technical term, it is not a semimartingale), no risk-neutral measure exists, and the apparatus of Chapter 15 onward has nothing to attach to. It is therefore not a candidate to replace (14.4). Mandelbrot proposed it for prices in the 1960s, with $H > 1/2$, and the proposal did not survive. The arbitrage is fragile in practice: Cheridito (2003) shows that a minimal delay between trades removes it, and Guasoni (2006) that proportional transaction costs of any size do, so the process is not absurd as a description of prices; it is merely useless as a pricing model. A "fractional Black–Scholes" formula exists and is sometimes quoted. It is not a price, its implied volatilities look nothing like a market surface, and no strategy replicates the option at that price; Section 8 of [TN21] explains all three.

Volatility is different. Spot volatility is not itself a traded asset, so no arbitrage argument rules out a rough path for it, and nothing forbids modeling it with a process that would be

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arbitrageable as a price. Gatheral, Jaisson, and Rosenbaum (2018) measured the roughness of realized volatility across many assets and found $H \approx 0.1$, far rougher than a Wiener path, and Section 21.8 shows what follows for option prices: the one number reproduces a feature of the volatility surface that two decades of classical models could not, and Example 14.8's arithmetic is the reason.

Simulating it. A Wiener path is simulated one independent increment at a time, and a fractional path cannot be, because the increments are not independent. The exact method writes down the covariance matrix (14.20) on the time grid, takes its Cholesky factor (the lower-triangular matrix L with $LL^\top$ equal to the covariance matrix), and multiplies a vector of standard normal draws by it, which is how Figure 14.5 was made, and why its three paths are the same randomness seen through three kernels. Each path costs the square of the number of steps and the factorization its cube, and Chapter 21's case study on rough volatility explains what replaced the method in production.

SUMMARY

A Markov process is one in which only the current value matters for the future. With independent changes over separate periods, variances add, so uncertainty grows with $\sqrt{T}$ rather than T. That is why volatility is scaled by $\sqrt{\Delta t}$ and why Chapter 13's tree used $u = e^{\sigma\sqrt{\Delta t}}$.

A Wiener process has increments $\Delta W = \varepsilon\sqrt{\Delta t}$ with ε standard normal and non-overlapping increments independent, so $W(T) - W(0) \sim N(0, T)$. Its paths are continuous everywhere and differentiable nowhere, which is why a different calculus is required.

A generalized Wiener process adds a constant drift and variance rate, $dx = a\,dt + b\,dW$; an Itô process lets both depend on the state and time.

Stock prices are modeled by geometric Brownian motion, $dS = \mu S\,dt + \sigma S\,dW$, because returns rather than dollar changes are what scale with the price level, and because the proportional form keeps the price positive. Over short intervals the noise term dominates the drift by an order of magnitude, which is why μ is hard to estimate; the hedge of Section 14.8 removes it from pricing.

Itô's lemma gives the process followed by a function $f(S, t)$. It differs from ordinary calculus by one term, $\frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2\,dt$, which survives because $dW^2 = dt$. That term is proportional to convexity and reappears as gamma in Chapter 18 and as the convexity adjustment in Chapter 30.

Applying the lemma to $\ln S$ gives $d\ln S = (\mu - \sigma^2/2)dt + \sigma\,dW$, so $\ln S_T$ is normal and S_T is lognormal. The expected price grows at μ while the median grows only at $\mu - \sigma^2/2$, so the gap widens with volatility and a volatile stock finishes below its own expected value more often than not. Confusing the two rates produces real and common errors in return reporting and product design.

Because a derivative and its stock are driven by the same dW, a position long $\partial f/\partial S$ shares and short the derivative carries no randomness, and μ cancels with it: Chapter 13's hedge in continuous time. Under the risk-neutral measure only the drift changes, from μ to r; the volatility carries over and the $-\sigma^2/2$ survives.

The model assumes constant volatility, continuous paths, independent returns, and known parameters. All four fail; knowing which one a given phenomenon violates identifies where in Parts III and IV its repair is found. Fractional Brownian motion is the standard process without independent increments: its variance over T is T^{2H}, so variances no longer add, and below $H = 1/2$ its paths are rougher than a Wiener path. It cannot model a traded price without admitting arbitrage, but it can model volatility, which Chapter 21 does with H far below $1/2$.

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FURTHER READING

The derivation of Itô's lemma is in [TN14], reproduced in Appendix A, which also makes the $dW^2 = dt$ argument precise. Beyond that, Neftci's introduction to the mathematics of financial derivatives bridges this level and a rigorous treatment, and Baxter and Rennie's account is a short exposition of why the change of measure works. Karatzas and Shreve's *Brownian Motion and Stochastic Calculus* is the standard graduate text on stochastic calculus, for a reader who wants the construction of the Itô integral itself rather than the rule that follows from it. For the empirical failures of geometric Brownian motion, Chapter 19's references are the place to start, since they organize the evidence around what the option market prices. Fractional Brownian motion entered finance through Mandelbrot and Van Ness (1968), who introduced the process and the Hurst parameter; Rogers (1997) is the short proof that it admits arbitrage as a price process, and Cheridito (2003) and Guasoni (2006) show the arbitrage vanishing under a minimal trading delay and under transaction costs respectively; the literature on the fractional Black–Scholes formula is cited in [TN21]. Gatheral, Jaisson, and Rosenbaum (2018) is the paper that found the process, with $H \approx 0.1$, in volatility.

PRACTICE QUESTIONS

14.1 A variable's change over one year has standard deviation 5. What is the standard deviation of its change over three years, and over one month? State the property you used.

14.2 A variable follows a generalized Wiener process with drift 3 per year and variance rate 16 per year, starting at 25. Give its distribution after two years and compute the probability that it is negative.

14.3 Explain why $dS = \mu\,dt + \sigma\,dW$ is unsuitable as a model for a stock price, giving two distinct reasons.

14.4 A stock at \$80 has $\mu = 12\%$ and $\sigma = 25\%$. Give the distribution of the proportional change over two weeks, and compare the sizes of the drift and noise terms.

14.5 Using Itô's lemma, find the process followed by $G = S^2$ when S follows (14.4).

14.6 A stock at \$60 has $\mu = 14\%$ and $\sigma = 35\%$. Compute the expected price and the median price after two years, and explain why they differ.

14.7 For the stock in the previous question, find the range within which the price after two years lies with 95% probability.

14.8 Explain in words why the term $\frac{1}{2}\sigma^2 S^2\,\partial^2 f/\partial S^2$ appears in Itô's lemma but not in the ordinary chain rule.

14.9 State the Markov property and explain why it is an assumption about information rather than about volatility. Explain what weak-form market efficiency has to do with it, and give one observed feature of returns that sits uneasily with it.

14.10 A stock has expected return μ. Explain why the expected continuously compounded return per year is $\mu - \sigma^2/2$ rather than μ, and state which of the two a manager quoting the arithmetic average of past annual returns is reporting.

14.11 Explain why Itô's lemma delivers a hedge and not merely a change-of-variable formula. State which term in the lemma the hedge removes, which term survives it, and why the surviving term is what the option is paid for.

14.12 State the risk-neutral process for a stock price. Explain what changes and what does not when moving to it from the real-world process, and why the volatility is untouched while the drift is not.

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FURTHER QUESTIONS

14.13 Example 14.6 found that a stock finishes below its expected value 58% of the time. Derive a general expression for $\Pr(S_T > \mathbb{E}[S_T])$ in terms of σ and T alone, show that it does not depend on μ, and evaluate it for $\sigma\sqrt{T} = 0.2$, 0.6, and 1.0. What happens as volatility rises?

14.14 A fund reports the arithmetic average of its annual returns over ten years as 12%. Explain what an investor's compounded return would have been if the volatility was 30%, state which of the two figures is the relevant one for an investor, and identify the circumstances in which the gap is largest.

14.15 Apply Itô's lemma to the forward price when the underlying pays a continuous dividend yield q, and show that under risk-neutral valuation the forward's drift is again zero. Explain the connection to Section 13.10.

14.16 Section 14.12 notes that volatility clustering violates the Markov property. Construct an argument that this does *not* contradict weak-form market efficiency, and identify precisely what a trader could and could not profit from if returns were unpredictable but their magnitudes were not.

14.17 Section 14.13 states that a change in fractional Brownian motion over any interval of length T has variance T^{2H}. Extend the argument that gives the correlation $2^{2H-1} - 1$ between consecutive yearly increments to the first year's increment and the increment over year $k+1$, and show that the correlation is $\tfrac{1}{2}\left[(k+1)^{2H} - 2k^{2H} + (k-1)^{2H}\right]$. Evaluate it for $k = 1, 5, 20$ at $H = 0.9$, and explain why a decay of order k^{2H-2}, slower than exponential, is the long memory the section names.

14.18 The Itô correction term is proportional to $\partial^2 f / \partial S^2$. Using this, explain without calculation whether the correction raises or lowers the drift of a long call position, and connect your answer to the statement in Chapter 12 that long-volatility positions are hurt by the passage of time.

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